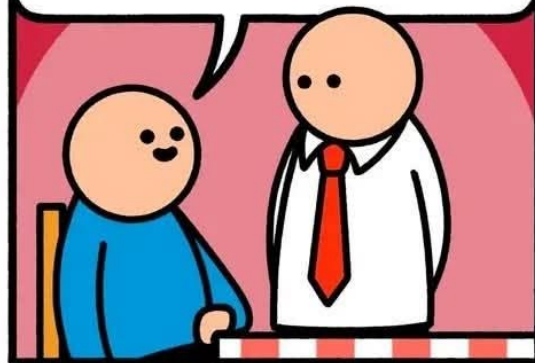


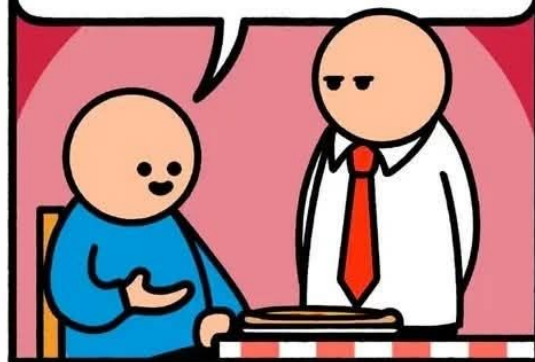
TWONKS

AI
DESIGN
LOGIC

GET ME A
CHEESE PIZZA



WOW, LOOK
WHAT I MADE!



LLM evaluation of NeurIPS submissions

David M.J. Tax

The BSc research question

- What is the use of Large Language Models in reviewing papers?
- Use NeurIPS 2024 papers, the NeurIPS paper checklist, and manual annotation
- Five BSc students focussed on different aspects

with the quality of a scientific paper. We are interested in seeing whether doing these checks using LLMs is trustworthy, and whether there are specific parts of the checklist the LLMs are bad at interpreting. This project will require you to also fill out the NeurIPS checklist by hand for about 20 papers.

Research Questions for the Sub-Projects

The overarching question is "Can an LLM provide quantitative values for the quality of a paper, using the NeurIPS Checklist mentioned above?". For this, some LLM has to be chosen, and it will require you to also fill out the NeurIPS checklist by hand for about 20 papers (to provide some ground truth annotations). In a combined effort with the whole group, data is annotated, some LLM's are compared and some evaluation is implemented.

Next, LLMs are used to answer how well a publication adhered to some points of the checklist:

1. How well can the LLM describe the claims that are made?
2. How well can the LLM list the limitations that are in their approach?
3. How well can the LLM derive the reproducibility of the proposed approach?
4. Can it be derived if the publication is an empirical (experimental) result, or a more theoretical result?
5. Can an LLM give an estimate of the significance of the result?



NeurIPS checklist

- Claims
- Limitations
- Theory, assumptions, proofs
- Experimental reproducibility
- Open access to data and code
- Experimental setting/details
- Statistical significance
- Compute resource
- Code of ethics
- Broader impact
- Safeguards
- Licenses
- Assets
- Crowdsourcing and research with human subjects
- Declaration of LLM usage



NeurIPS Paper Checklist Guide

The NeurIPS Paper Checklist is designed to encourage best practices for responsible research, addressing issues of reproducibility, transparency, research ethics, and robustness. It is included in [the LaTeX style file](#). Do not remove the checklist: **The papers not following the checklist will be desk rejected.**

All submissions must be in PDF format, and in a single PDF file include, in this order: (1) the main paper, (2) optional technical appendices that support the paper with additional proofs or data, and (3) the NeurIPS paper checklist. The checklist should therefore follow the submission of supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer each question in the checklist:

- You should answer yes, no, or n/a.
- NA means either that the question is Not Applicable for that particular paper or that the answer is Not Available.
- If you would like to explain your answer (e.g., explain why the answer is "no" or "n/a" to the reviewers), consider referencing the section(s) of the paper that provide the justification.
- You can also optionally write a short (1–2 sentence) justification of your answer.

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will be asked to also include the checklist with the final version of your paper, and its final version will be published with the paper. **answers are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers.**

Reviewers will be asked to use the checklist as one of the factors in their evaluation. While "no" is preferable to "no", it is perfectly acceptable to answer "no" provided a proper justification is given (e.g., "no because the dataset is not publicly available" or "no because the dataset license for the dataset we used"). **In general, answering "no" or "n/a" is not good enough.** If questions are phrased in a binary way, we acknowledge that the true answer is often not binary. Just use your best judgement and write a justification to elaborate. All supporting material should be included in the **main paper or the supplemental material provided in the appendix**. If you answer "no" or "n/a" the justification please point to the section(s) where related material for the question is provided.

We provide guidance on how to answer each question below. You may additionally

The five questions

- A. How well can an LLM distinguish between empirical and theoretical papers? Or answer all items in the checklist?
- B. How well can an LLM extract the scientific claims made in a paper?
- C. How well can the limitations (as described in the paper) be detected?
- D. How well can an LLM judge the reproducibility of a paper?
- E. How well can an LMM assess the experimental statistical significance of the results?



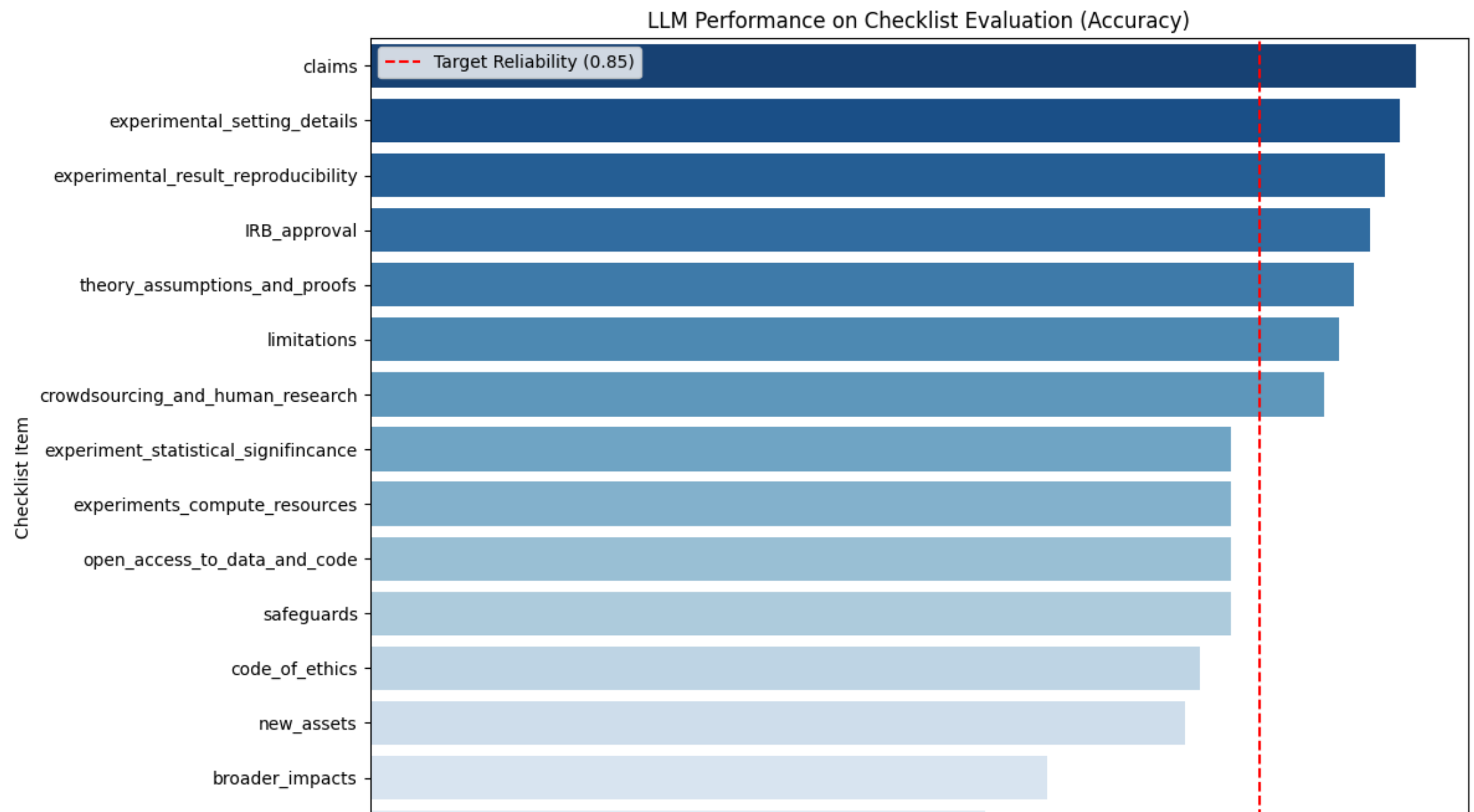
The datasets and models

- Around 80 papers, 5 annotators
- For some subsequent questions ('How well can implicit limitations be detected?'), subsets of the papers are annotated (like 30)
- For checking claims, an adversarial dataset is created
- For checking reproducibility, an alternative dataset is used (the Machine Learning Reproducibility Challenge)
- LLM: Gemini 3.0 Flash



Some results

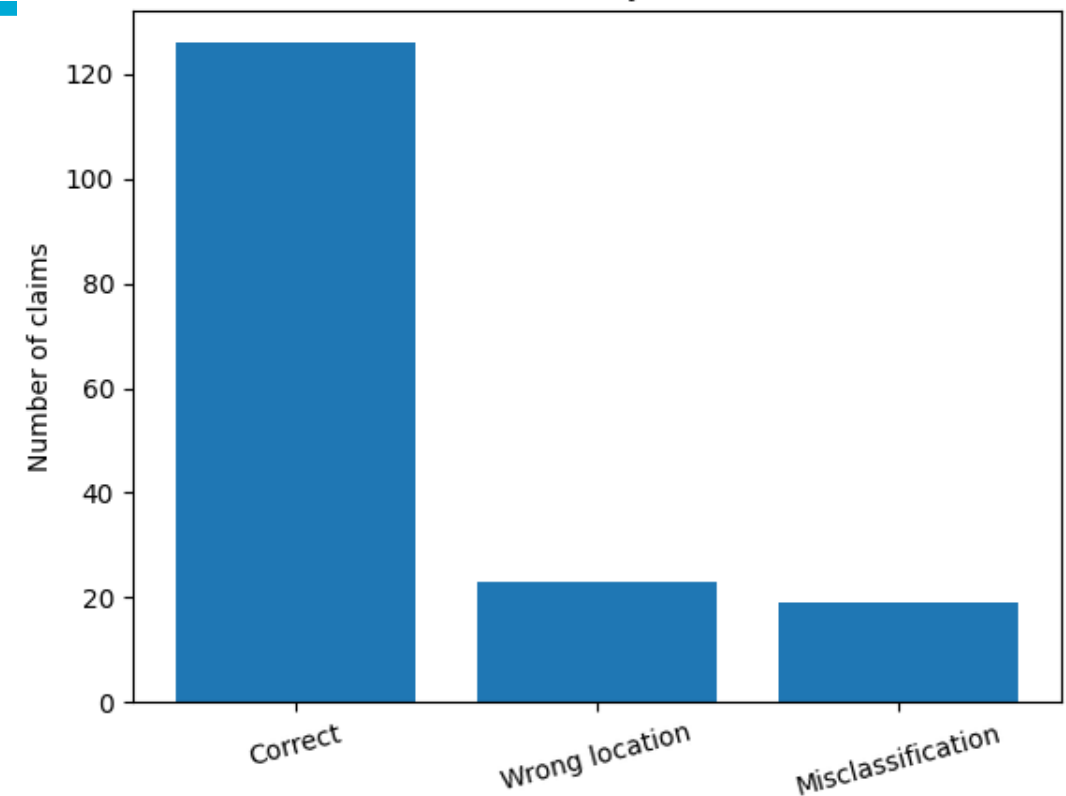
- To distinguish between empirical-theoretical-hybrid is easy (all correct). Easy.
- To determine the 16 checklist items, depends on how explicit the definition is:



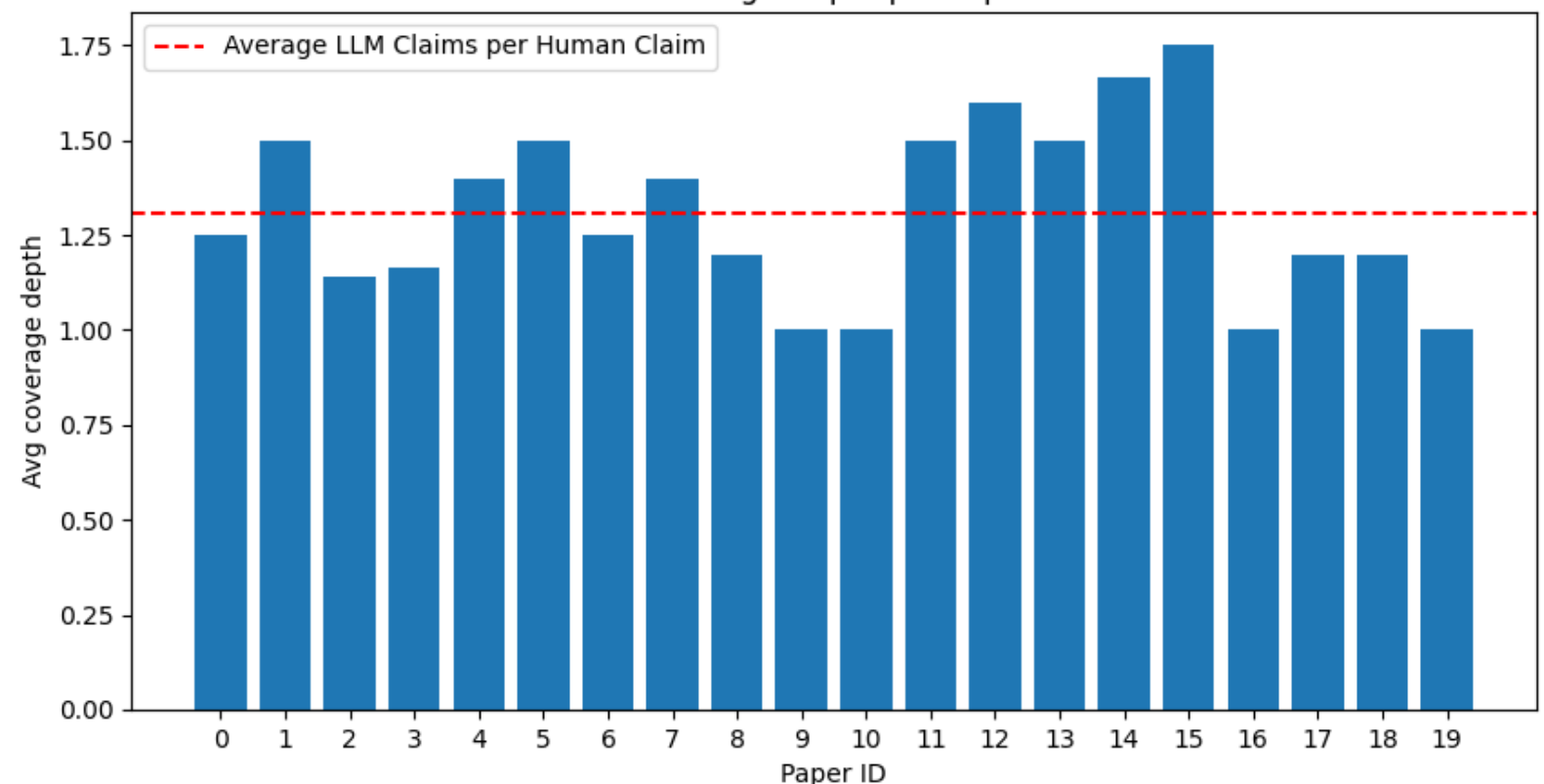
How well are claims detected?

- 75% of the 168 LLM-extracted claims are correct
- Some claims were statements from Methods, Experiments, Results sections
- Also some 'oversegmentation' happens

LLM Claim Quality Breakdown



Coverage Depth per Paper



Can limitations be detected?

- From the 137 ground truth limitations, 174 were detected (oversegmentation)
- The explicit section 'Limitations' from the paper is used.

Textual Limitation Extraction Confusion Matrix
Total Ground Truth Limitations: 137 | Total LLM Extracted: 174

Actual Positive (Ground Truth Limitation)	True Positives (TP) Exact Matches: 106 Differently Chunked: 18 (Successfully extracted)	False Negatives (FN) Count: 13 (Missed by LLM)
	False Positives (FP) Count: 50 (Hallucinated / Extra)	True Negatives (TN) N/A (Not applicable for text extraction)
Actual Negative (Not a Limitation)	Predicted Positive (Extracted by LLM)	Predicted Negative (Missed by LLM)



Can limitations be detected?

- Then the explicit section 'Limitations' is removed from the paper.
- From the 17 ground truth limitations, 16 were detected
- Hmmmm

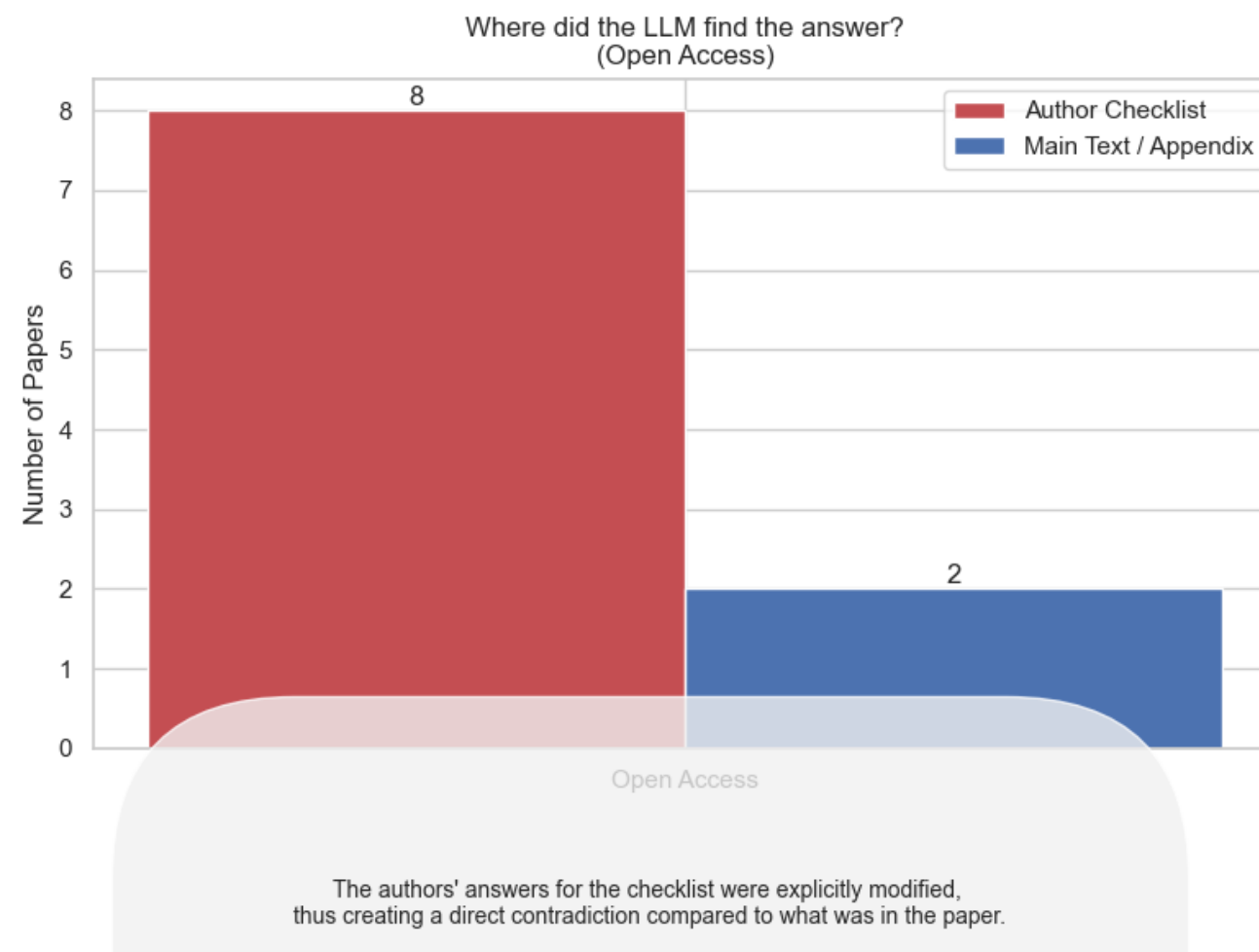
Textual Limitation Extraction Confusion Matrix
Total Ground Truth Limitations: 17 | Total LLM Extracted: 16

Actual Positive (Ground Truth Limitation)	True Positives (TP) Exact Matches: 8 Differently Chunked: 4 (Successfully extracted)	False Negatives (FN) Count: 5 (Missed by LLM)
	False Positives (FP) Count: 5 (Hallucinated / Extra)	True Negatives (TN) N/A (Not applicable for text extraction)
Actual Negative (Not a Limitation)	Predicted Positive (Extracted by LLM)	Predicted Negative (Missed by LLM)



Reproducibility

- 30 adversarial test cases where non-valid repository links were introduced, compute resources were not mentioned, but where model specifications were correct
- Also asked evidence for the answer



More results

- Prompt for statistical significance:

- This resulted in many false negatives ('too strict', according to the students)
- A looser prompt was created

B.1 Prompt 1

You are an expert peer reviewer for the NeurIPS conference. You will be provided with a submitted paper and must answer the following question using one of the following values only: "yes", "no", or "na".

You should also provide a brief justification for your answer. In the justification, explicitly refer to the relevant section(s) of the paper that support your assessment. Your judgment must be based solely on information explicitly stated in the PDF.

Question Does the paper report error bars suitably and correctly define them, or provide other appropriate information regarding the statistical significance of the experiments?

Guidelines

- The answer "na" should be used if the paper does not include experiments.
- The answer "yes" should be given if the reported results include error bars, confidence intervals, or statistical significance tests, at least for the experiments supporting the main claims of the paper.
- The factors of variability captured by the error bars should be clearly specified (e.g., train/test split, initialization, random parameter sampling, or complete experimental runs under fixed conditions).
- The method used to calculate the error bars should be explained (e.g., analytical formula, library implementation, bootstrap procedure, etc.).
- Any assumptions underlying the calculations should be stated (e.g., normally distributed errors).
- It should be clearly indicated whether the reported error bars correspond to the standard deviation or the standard error of the mean.
- Reporting one-standard-deviation (1σ) error bars is acceptable, provided this is explicitly stated. Reporting standard error of the mean (SEM) is also acceptable, provided this is explicitly stated.



Statistical significance

- More lenient prompt:
- “Encouraged a more holistic evaluation of the paper and reduced the tendency of the model to classify papers as non-compliant.”

B.3 Prompt 3

You are assessing a NeurIPS submission. You will be provided with a paper and must determine whether the paper reports error bars, confidence intervals, statistical significance tests, or other appropriate measures of experimental uncertainty. Make a best-effort judgment based solely on the evidence presented in the paper. Do not require perfect reporting. If the paper provides meaningful uncertainty estimates for the main experimental results, answer "yes" even if some methodological details are missing.

Question Does the paper report error bars suitably and correctly define them, or provide other appropriate information regarding the statistical significance of the experiments?

Decision Rules

- Answer "na" if the paper contains no experiments.
- Answer "yes" if the paper reports error bars, confidence intervals, statistical significance tests, results across multiple runs, or other reasonable measures of uncertainty for the main experimental claims.
- Answer "yes" even if assumptions, formulas, or implementation details are not fully described.
- Answer "no" only when there is little or no evidence that experimental uncertainty or statistical significance information has been reported.

Helpful Indicators The following are useful indicators but should not be treated as strict requirements:

- Error bars in figures or tables.
- Confidence intervals.
- Statistical significance tests.
- Standard deviations or standard errors.
- Results reported across multiple random seeds or experimental runs.
- Discussion of variability, robustness, or uncertainty.

Some quotes

- “These observations suggest that the current limitations of the model arise less from retrieving information and more from interpreting scientific context.”
- “In a paper which contained no formal limitations according to the ground truth, the model hallucinated an extraction by pulling a standard result-reporting sentence ...”
- “...performance decreases for more subjective criteria that require interpretation or external context.”



Obvious conclusions

- If it is explicitly in the text (claims, proofs, conclusions) it is easily found
- Internal consistency, correctness of reasoning, logic, broader limitations are much, much worse.



Less obvious conclusion

Author



submit



Reviewer



- Classic review process



Less obvious conclusion

Author

Reviewer



review

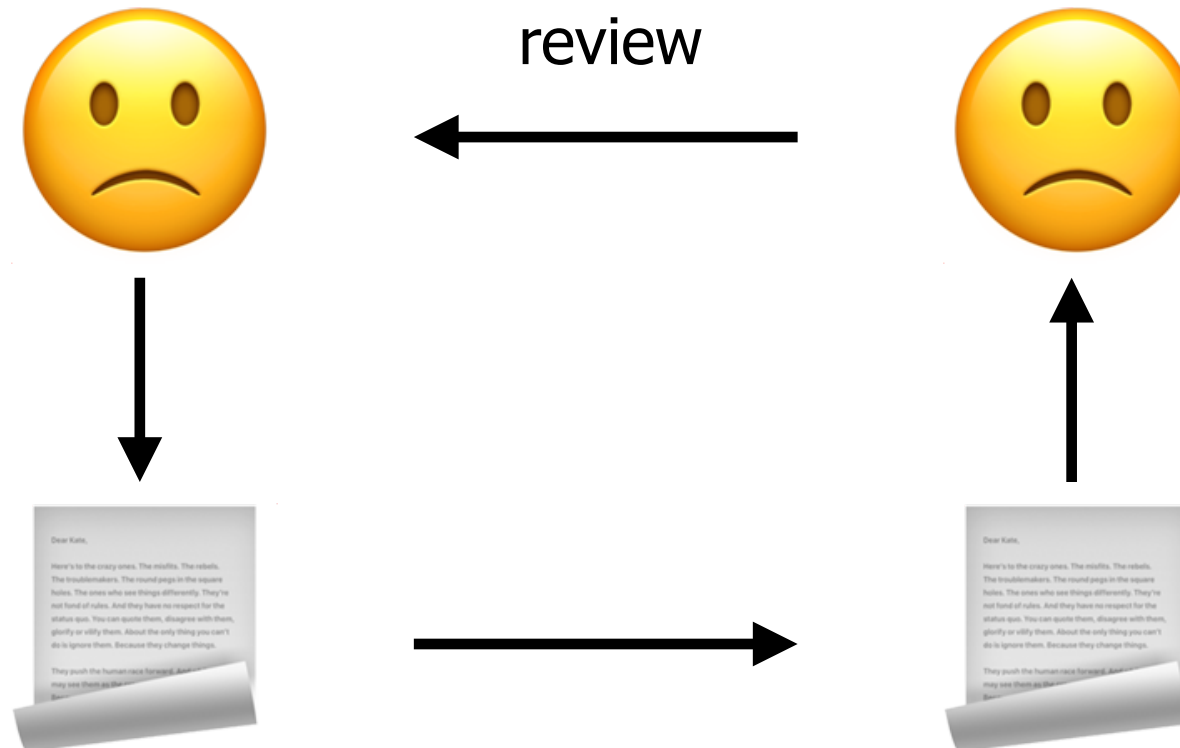


- Classic review process



Author

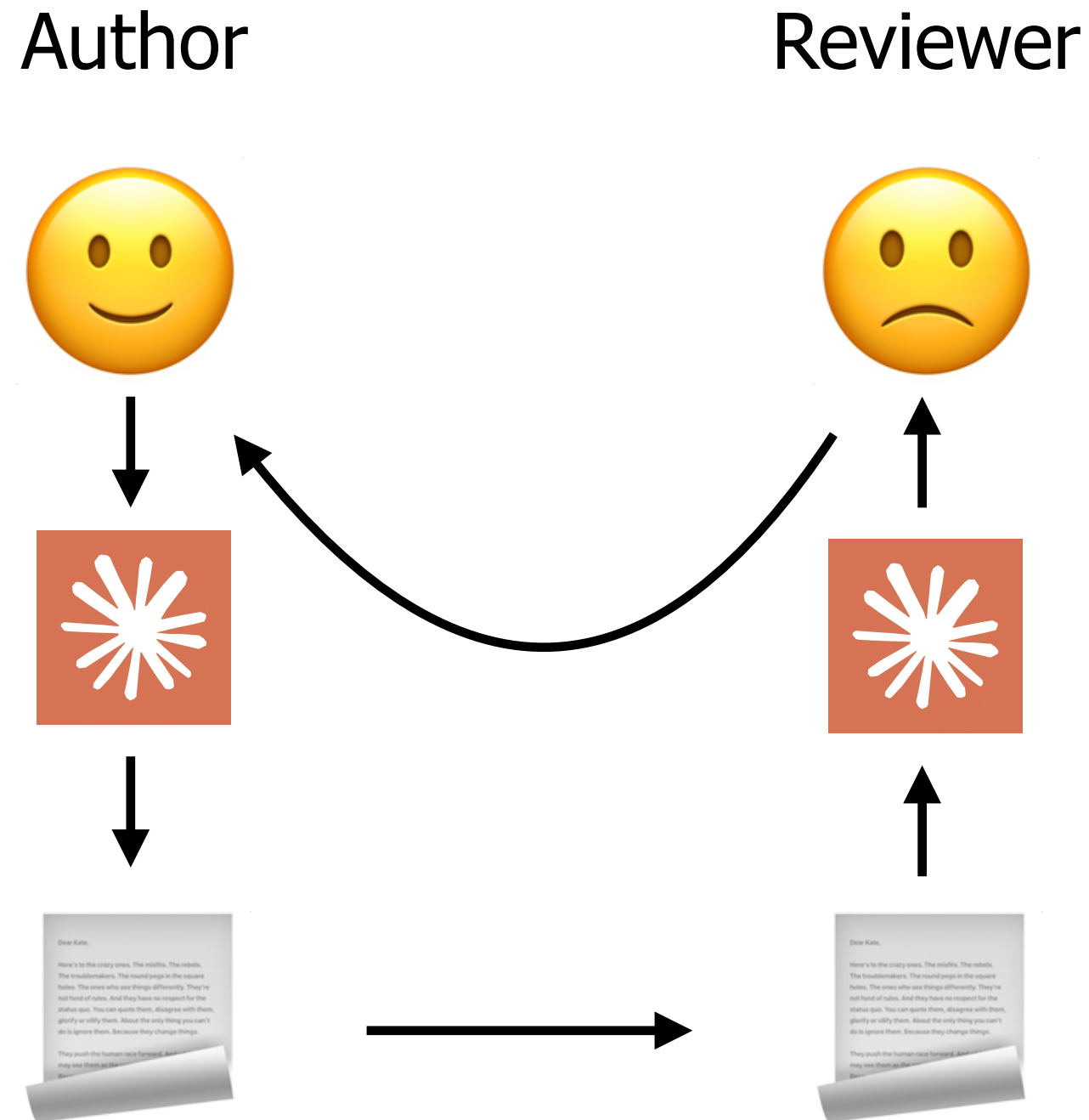
Reviewer



- Classic review process



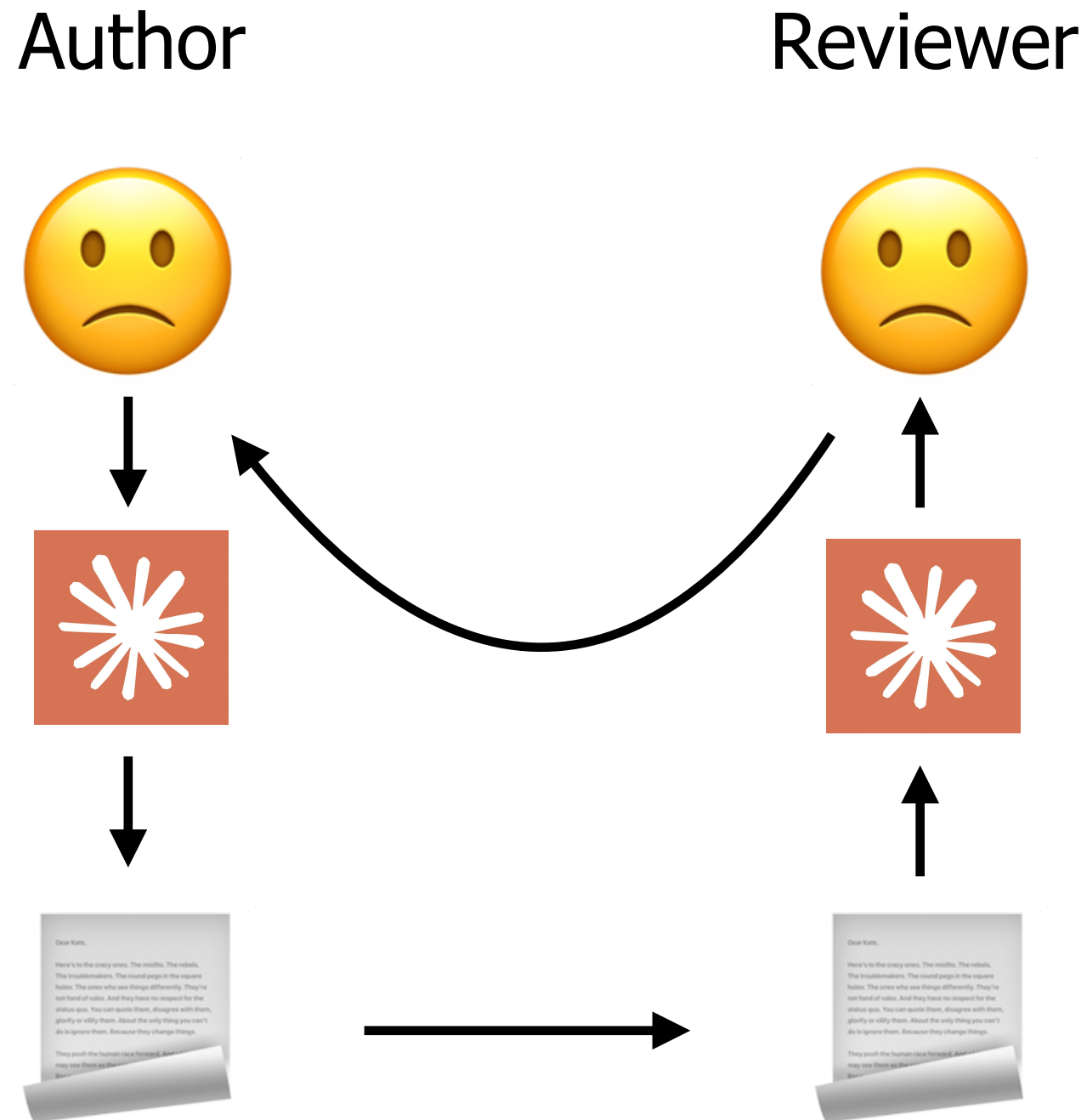
Influence of LLM



- Modern review process



Influence of LLM



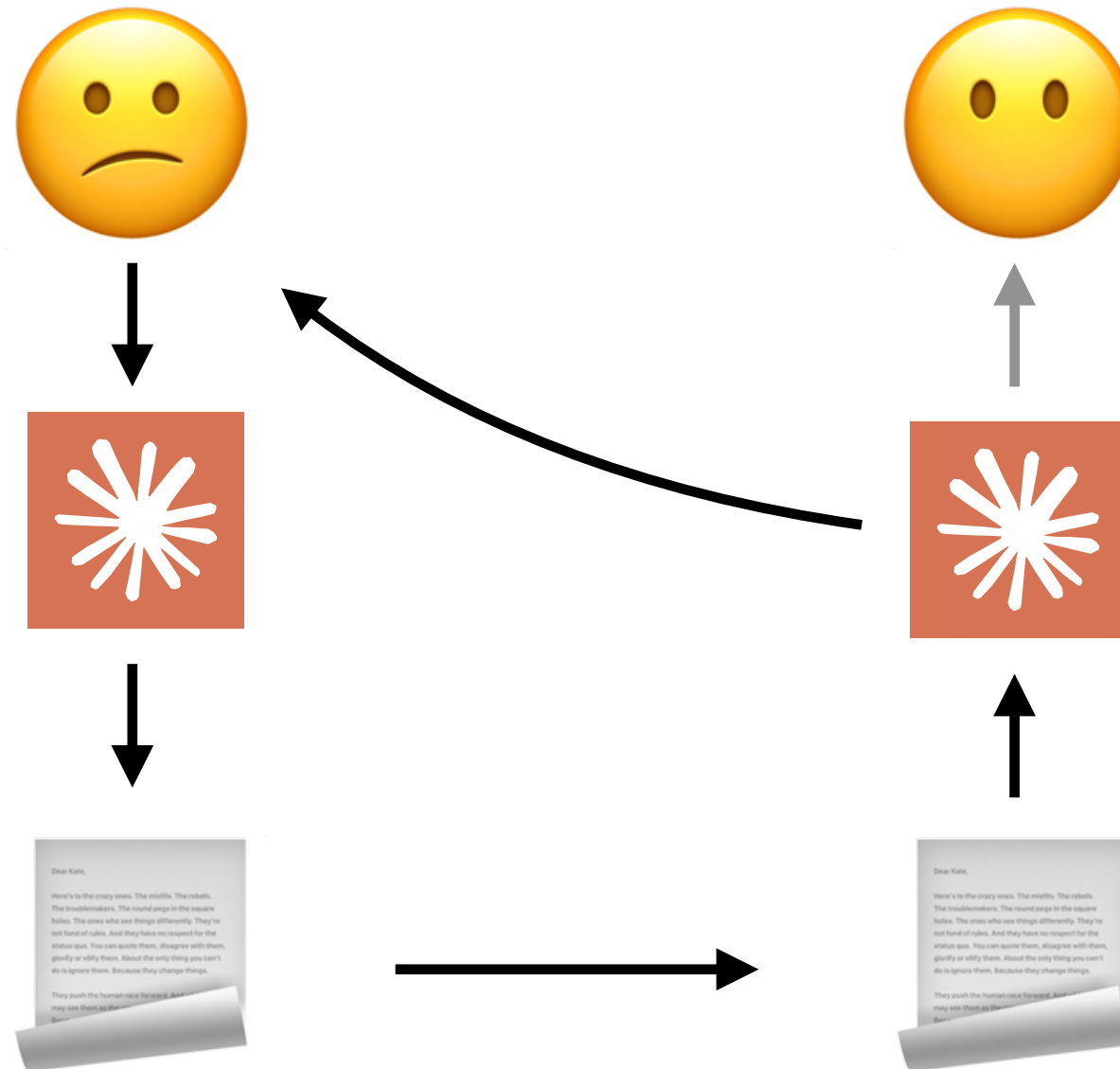
- Modern review process



Influence of LLM

Author

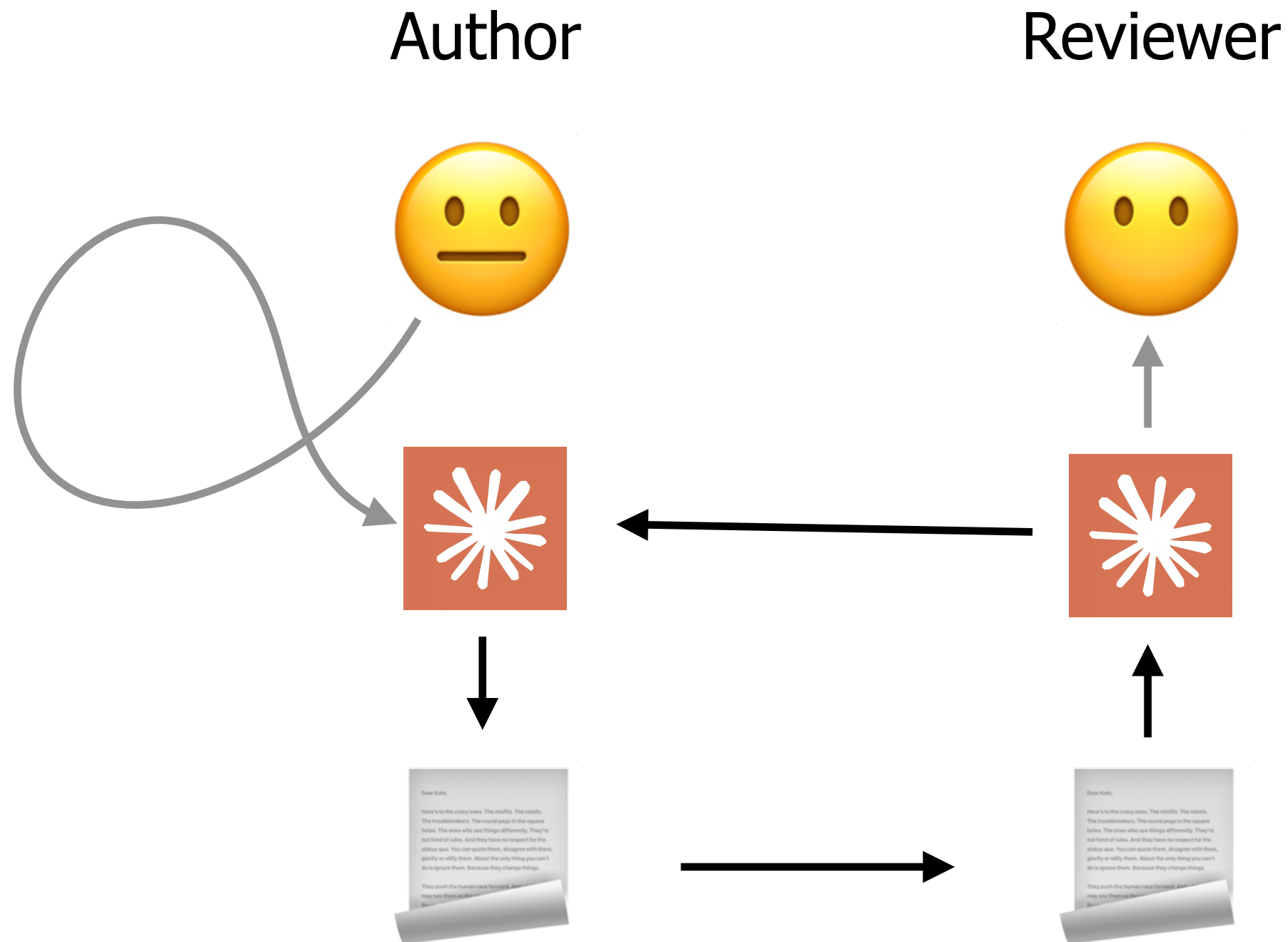
Reviewer



- Modern review process



Influence of LLM



- Modern review process





Question of conscience

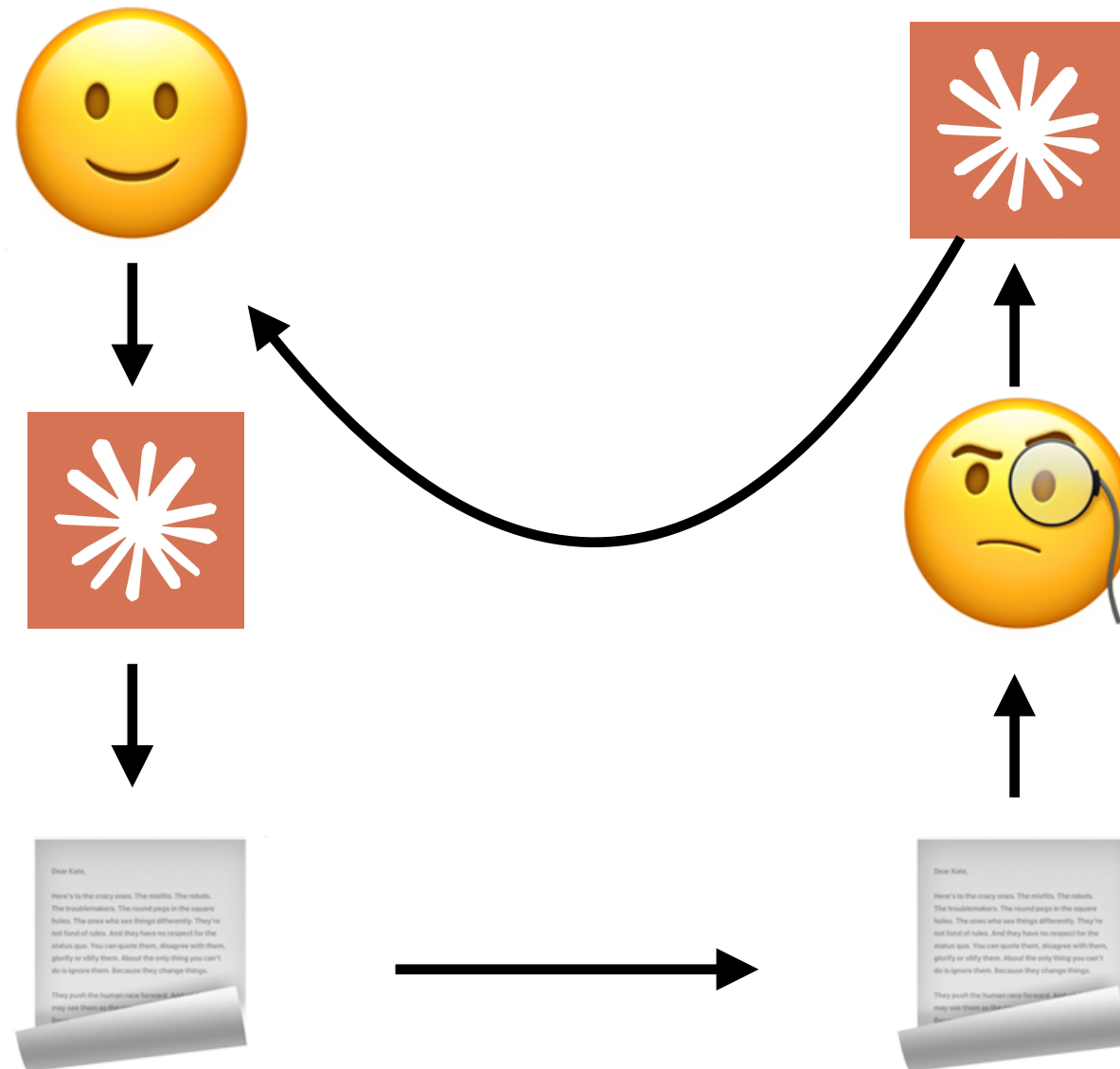
- Do you add a 'list of contributions', or list of 'research questions' to the introduction section of your paper?
- How many of you do teach students to do this?
- Why?



Good use of LLM

Author

Reviewer



- Solution?



Conclusions

- The real issue:
- Detecting the consistency/logic/limitations/... **as written in the paper**
VS
detecting the **real** consistency/logic/limitations/... in the paper
- People/researchers themselves start behaving as LLMs, and forget to think

